Problem

The problem I want to address is the quality and naturalness of Text-to-Speech (TTS) systems used in common Augmentative and Alternative Communication (AAC) software.

Why It Matters

Many AAC systems rely on synthesized speech to allow users to communicate an essentially unlimited range of words and sentences. However, synthesized voices can sometimes produce incorrect or unnatural pronunciations, particularly for less common words, names, or context-dependent pronunciations. The voices may also sound noticeably different from natural human speech. For some children they may never be able to fully communicate without the help of a device, but many have their speech development supported by the use of an AAC. This is why it's important to make sure that those pronunciations are correct, because it is one of the main facets in which they learn to speak themselves.

While some AAC systems address pronunciation and naturalness by using pre-recorded speech, pre-recorded messages limit the number of things a child can communicate and make it difficult to generate new, complex sentences. So, it works better for younger children who will communicate with more simplicity, but becomes less viable as the child gets older.

This creates a tradeoff between flexibility and natural-sounding speech. Synthesized speech provides the flexibility to say virtually anything, while pre-recorded speech can sound more natural but is limited to previously recorded words or phrases.

Proposed Solution

My proposed solution is to use deep learning to develop a more natural-sounding TTS system specifically for AAC applications. The goal would be to preserve the flexibility of synthesized speech—allowing a child to generate virtually any sentence—while producing speech that is more natural and human-like.